

MODEL RESEARCH

Model 1: Statistical Analysis Model

Model Overview

The Statistical Analysis Model is used to analyze and summarize cyber threat data in a structured and quantitative manner. This model focuses on calculating basic statistical measures such as frequency, count, percentage, and distribution of cyber incidents.

Relevance to the Dashboard

In the Interactive Cyber Threat Visualization Dashboard, this model generates key summary insights that are visually displayed. It supports dashboard components such as:

- Total number of cyber incidents
- Attack type distribution
- Severity-wise breakdown
- Most targeted systems and regions

These insights allow users to quickly understand the overall cybersecurity posture without examining raw data.

Working Mechanism

The model processes raw cyber incident records and applies aggregation techniques such as grouping and counting. The processed results are visualized using charts like bar graphs, pie charts, and tables, making complex datasets easy to interpret.

Advantages

- Easy to implement and computationally efficient
- Provides quick and clear insights
- Supports effective visualization
- Helps establish baseline cyber activity behavior
- Useful for both technical and non-technical users

Limitations

- Does not predict future attacks
- Cannot detect unknown threats independently
- Limited to historical data analysis
- Not adaptive to evolving attack patterns

Applicability to the Project

This model forms the foundation of the dashboard by enabling basic cyber threat analysis and visualization. It supports decision-making by highlighting high-risk attack categories and vulnerable systems.

Model 2: Time-Series Analysis Model

Model Overview

The Time-Series Analysis Model analyzes cyber threat data based on time to understand how attacks vary across daily, weekly, or monthly intervals.

Relevance to the Dashboard

This model enables the dashboard to display:

- Attack trends over time
- Peak attack periods
- Sudden spikes or drops in attack frequency

These insights are visualized using line charts and trend graphs.

Working Mechanism

Cyber incidents are organized using timestamps and grouped into time intervals. The model identifies trends, seasonality, and abnormal fluctuations, which are visualized to provide a temporal understanding of cyber threats.

Advantages

- Identifies trends and seasonal patterns
- Detects sudden increases in attacks
- Supports proactive monitoring
- Easy to integrate with dashboards
- Enhances situational awareness

Limitations

- Does not explain attack causes
- Limited predictive capability without forecasting models
- Sensitive to incorrect timestamp data
- Requires sufficient historical data

Applicability to the Project

This model helps visualize attack trends and identify abnormal activity periods, aiding in better planning and risk assessment.

Model 3: Naive Bayes Classification Model

Model Overview

The Naive Bayes Classification Model is a probabilistic machine learning model based on Bayes' theorem. It is widely used for classifying cyber attacks and threat severity levels.

Relevance to the Dashboard

The dashboard visualizes outputs such as:

- Attack type classification
- Severity prediction
- Distribution of predicted threat categories

This adds intelligent insights beyond historical data.

Working Mechanism

The model calculates the probability of an incident belonging to a specific class using historical labeled data. The class with the highest probability is assigned and visualized.

Advantages

- Fast and computationally efficient
- Works well with large datasets
- Effective for categorical data
- Easy to interpret

Limitations

- Assumes feature independence
- Less accurate for complex patterns
- Sensitive to training data quality

Applicability to the Project

This model enables automated threat categorization, improving dashboard intelligence and analytical depth.

Model 4: Random Forest Classification Model

Model Overview

Random Forest is an ensemble learning model that combines multiple decision trees to improve classification accuracy.

Relevance to the Dashboard

The dashboard displays:

- Predicted risk levels
- Severity classification
- System risk ranking

Working Mechanism

Multiple decision trees are trained on different data subsets, and predictions are combined using majority voting to reduce overfitting.

Advantages

- High classification accuracy
- Handles complex datasets well
- Reduces overfitting
- Suitable for cybersecurity data

Limitations

- Less interpretable
- Higher computational cost
- Longer training time

Applicability to the Project

This model strengthens predictive threat analysis and enhances risk prioritization in the dashboard.

Model 5: Anomaly Detection Model

Model Overview

The Anomaly Detection Model identifies unusual cyber activities that deviate from normal behavior.

Relevance to the Dashboard

It highlights:

- Sudden attack spikes

- Unusual severity patterns
- Abnormal system behavior

Working Mechanism

The model learns baseline behavior and flags deviations using techniques such as Z-score analysis or Isolation Forests.

Advantages

- Detects zero-day attacks
- Does not require labeled data
- Supports early detection

Limitations

- Possible false positives
- Requires threshold tuning
- Does not classify attacks

Applicability to the Project

This model enhances dashboard alerting by identifying suspicious activities requiring immediate attention.

Model 6: Clustering Model

Model Overview

The Clustering Model groups similar cyber incidents without using labeled data.

Relevance to the Dashboard

It visualizes:

- Similar attack behavior groups
- Emerging attack patterns

Working Mechanism

Algorithms such as K-Means or DBSCAN cluster incidents based on similarity.

Advantages

- Discovers hidden patterns
- Reduces data complexity

- Useful for exploratory analysis

Limitations

- Difficult cluster interpretation
- Sensitive to parameter selection

Applicability to the Project

This model helps analyze attack behaviors and supports strategic threat grouping.

Model 7: Frequency-Based Risk Scoring Model

Model Overview

This model assigns risk scores based on attack frequency and severity.

Relevance to the Dashboard

It displays:

- High-risk systems
- High-risk regions
- Risk rankings

Working Mechanism

Risk scores are calculated using attack frequency and severity metrics and visualized through heat maps or rankings.

Advantages

- Simple and intuitive
- Supports prioritization
- Easy dashboard integration

Limitations

- Ignores attack sophistication
- Depends on historical data

Applicability to the Project

This model enables effective resource allocation and proactive mitigation.

Model 8: Rule-Based Detection Model

Model Overview

The Rule-Based Detection Model identifies known threats using predefined rules.

Relevance to the Dashboard

It supports:

- Real-time alerts
- Rule violation summaries

Working Mechanism

Incidents matching predefined conditions are flagged and visualized.

Advantages

- Easy to interpret
- Low computational cost
- Effective for known threats

Limitations

- Cannot detect unknown attacks
- Requires continuous updates

Applicability to the Project

This model complements ML models by reliably detecting known attack patterns.

Model 9: Visualization-Driven Analytical Model

Model Overview

This model emphasizes insight generation through interactive visual analysis.

Relevance to the Dashboard

It enables:

- Maps, charts, treemaps
- Interactive filtering and drill-down

Working Mechanism

Processed data is converted into interactive visuals for human-centric analysis.

Advantages

- Improves understanding
- Supports faster decisions
- User-friendly

Limitations

- Depends on data quality
- Requires careful design

Applicability to the Project

This model is the **core of the dashboard**, integrating all insights visually.

Model 10: Hybrid Model

Model Overview

The Hybrid Model integrates statistical, machine learning, anomaly detection, and visualization techniques.

Relevance to the Dashboard

It simultaneously displays:

- Statistical summaries
- Predictions
- Anomaly alerts
- Risk scores

Working Mechanism

Multiple models operate in layers, and their outputs are combined and visualized.

Advantages

- High accuracy
- Comprehensive analysis
- Reduced false positives

Limitations

- Complex implementation

- Higher resource requirements

Applicability to the Project

This model provides the most comprehensive cyber threat analysis for proactive security management.

Model Comparison Table

Model No	Model Name	Purpose	Nature	Dashboard Role
1	Statistical Analysis	Summarization	Analytical	Charts & metrics
2	Time-Series Analysis	Trend analysis	Analytical	Trend graphs
3	Naive Bayes	Classification	ML	Attack categorization
4	Random Forest	Risk prediction	ML	Risk scoring
5	Anomaly Detection	Detect anomalies	ML	Alerts
6	Clustering	Pattern grouping	ML	Behavior analysis
7	Risk Scoring	Risk ranking	Analytical	Prioritization
8	Rule-Based	Known threat detection	Rule-based	Alerts
9	Visualization-Driven	Visual analysis	Visualization	Dashboard UI
10	Hybrid Model	Combined analysis	Hybrid	Full threat insight

Conclusion

The integration of these models provides a comprehensive framework for cyber threat analysis and visualization. By combining analytical, machine learning, and visualization-driven approaches, the dashboard enables effective monitoring, detection, and decision-making, thereby strengthening proactive cybersecurity management.